Ahmed Loughzali

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LinkedIn
GitHub
Personal website

Education

Fall 2026 **Incoming Exchange Student.**
Gwangju Institute of Science and Technology (GIST), South Korea
Focusing on Robotics and Physical AI. Selected Coursework: Multiple View Geometry for Spatial AI, Robotics, Haptics, AI for Healthcare, Biomechanics.

2022 – 2027 **MSc in Computer Science.**
IMT Nord Europe, France
Majoring in AI & Data Science.

Experience

2026 – 2027 **Incoming AI Researcher.**
Algoverse, Remote
Selected for a highly competitive research program with a scholarship and compute grant. Aiming to publish at top-tier AI venues.

Apr. 2026 – Sept. 2026 **AI & Robotics Research Intern.**
Enchanted Tools, Paris, France
– Working on the brain of Mirokaïs, Europe's most advanced social humanoid robots. Implementing and benchmarking state-of-the-art speech processing and turn-taking models (VAP, TurnGPT, Semantic-VAD via Silero & custom Whisper).
– Optimizing models for real-time human-robot interaction under strict inference latency constraints, notably by applying model quantization.
– Streamlined the edge deployment lifecycle, handling cross-compilation, gRPC service integration, and CI/CD automation via GitLab pipelines.
– Selected to represent the company and pitch the robot's technical capabilities to enterprise customers at major industry events (VivaTech, Microsoft stand at Santexpo).

May 2025 – Aug. 2025 **AI Engineer Intern.**
Ferromobile, Paris, France
– Built and optimized a real-time Text-to-Speech service (FastAPI, Piper TTS) on NVIDIA Jetson Xavier, achieving $\sim$0.6s latency with caching, and integrated it via TCP/HTTP.
– Designed and deployed a local RAG pipeline (Docling, LangChain, ChromaDB, Mistral 7B via Ollama) with a Streamlit web interface for intelligent querying of technical documents.
– Containerized the full architecture using Docker Compose and Dev Containers to ensure reproducible and reliable deployment on embedded industrial hardware.

Projects & Research

Feb. 2026 **1st Place Winner, Mistral AI x GE Healthcare Hackathon.**
OncovIA
Developed a multimodal pipeline for oncology combining Computer Vision and Agentic RAG to help radiologists generate accurate, patient-specific follow-up reports.

Oct. 2025 – Feb. 2026 — **Drone Detection using Spiking Neural Networks.**
Research Project

Authored a review paper on drone detection focusing on neuromorphic computing and energy-efficient SNNs. Fine-tuned spiking neural network architectures (Spiking YOLOv8, Spiking SSD) on the FRED dataset. Achieved 92% accuracy with a 69M parameter model optimized for low-power edge inference.

Feb. 2026 — **3rd Place, Text to Motion Retrieval.**
Multimodal Kaggle Competition

Designed a parameter-efficient dual-encoder using Contrastive Learning (Symmetric InfoNCE) to retrieve 3D motion capture sequences from text during a 1 week sprint.

Sep. 2024 – Apr. 2025 — **Ozone Measurement System on Drone.**
Embedded Hardware & IoT Project

Designed and built a low-cost (< 150€) autonomous embedded system mounted on a drone for real-time ozone measurement (ESP8266, SEN0321, GPS). Handled full hardware design, wiring, C++ embedded software, and remote server networking (Flask + InfluxDB + Grafana), achieving an 80% correlation with a 3000€ professional reference sensor.

Technical Strengths

Languages	Python, C++, SQL, Bash, LaTeX
Frameworks	PyTorch, Hugging Face, LangChain, FastAPI, gRPC, OpenCV, NumPy, Pandas, Scikit-learn
Tools & Infra	Linux, Git, Docker, GitLab CI/CD
Hardware & Edge	NVIDIA Jetson, ESP, Cross-Compilation, Low-Power Inference

Languages

English	Full Professional Proficiency	*TOEIC: 980/990*
French	Mother tongue	
Arabic	Mother tongue	
Spanish	Intermediate	
Russian	Beginner	

Interests

Sports	Boxing, Wrestling, Hiking.
Other	Reading.